AQ2.6: Activity Questions 6 - Not Graded

This assignment will not be graded and is only for practice.
Level 1:

1 point
Let $Ax = b$Ax = b be a matrix representation of a system of linear equations. Let $[R|c]$[R|c] be the reduced row echelon of the augmented matrix $[A|b]$[A|b] corresponding to the system. Choose the set of correct options.

[Hint: Recall reduced row echelon form.]

- [ ] If the system $Rx = c$Rx = c has infinitely many solutions, then the system $Ax = b$Ax = b has infinite solutions.
- [ ] If the system $Rx = c$Rx = c has no solutions, then the system $Ax = b$Ax = b has a unique solution.
- [ ] If the system $Rx = c$Rx = c has a unique solution, then the system $Ax = b$Ax = b has no solution.
- [ ] If the system $Rx = c$Rx = c has a unique solution, then the system $Ax = b$Ax = b has a unique solution.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If the system $Rx = c$Rx = c has infinitely many solutions, then the system $Ax = b$Ax = b has infinite solutions.
If the system $Rx = c$Rx = c has a unique solution, then the system $Ax = b$Ax = b has a unique solution.

Consider a system of linear equations $\begin{matrix} 2x_1 + x_2 = 1 \\ -x_1 + x_3 + x_4 = -1 \\ x_1 + x_2 - x_3 + x_4 = 2 \\ -x_1 + x_3 + x_4 = 1. \end{matrix}$2x1 + x2 = 1−x1 + x3 + x4 = −1x1 + x2 − x3 + x4 = 2−x1 + x3 + x4 = 1.If the following matrix represents the augmented matrix of the system, then answer the questions 2,3 and 4.

Find the value of a_{22} a22.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point
Find the sum of the elements of the 4-th row of the augmented matrix.

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 2
1 point
Find the sum of the elements of the 3-rd column of the augmented matrix.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

1 point
Let $Ax = b$Ax = b be a matrix representation of a system of linear equations and b = 0b = 0. Choose the set of correct options.

[Hint: Recall the definition of the trivial solution of a system of linear equations and applicability of Gauss Elimination Method.]

- [ ] If AA is an invertible matrix then the system has no solution.
- [ ] If AA is an invertible matrix then the system has a unique solution.
- [ ] If AA is an invertible matrix then the trivial solution is the only solution for the system.
- [ ] If $det(A)$ = 0det(A) = 0, then the system has infinitely many solutions.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If AA is an invertible matrix then the system has a unique solution.
If AA is an invertible matrix then the trivial solution is the only solution for the system.
If $det(A)$ = 0det(A) = 0, then the system has infinitely many solutions.
1 point
Let $Ax = b$Ax = b be a matrix representation of a system of linear equations and $b \neq 0$b = 0. Choose the set of correct options.

- [ ] If AA is an invertible matrix, then the system has no solution.
- [ ] If AA is an invertible matrix, then the system has a unique solution.
- [ ] If AA is an invertible matrix, then the trivial solution is a solution for the system.
- [ ] If $det(A)$ = 0det(A) = 0, then either the system has no solution or the system has infinitely many solutions.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If AA is an invertible matrix, then the system has a unique solution.
If $det(A)$ = 0det(A) = 0, then either the system has no solution or the system has infinitely many solutions.
1 point
Consider the following systems of linear equations:

System I: $x - y = 3$, $-y + 2z = 1$, $x + y + z = 0$

System II: $x - y = 3$, $2y + z = 1$, $6y + 3z = 0$

System III: $x - y = 3$, $2y + z = 1$, $6y + 3z = 3$

System I:System II:System III:

x − y = 3−y + 2z = 1x + y + z = 0x − y = 32y + z = 16y + 3z = 0x − y = 32y + z = 16y + 3z = 3

Choose the correct option.

- ☐ All the three systems have a unique solution.
- ☐ System I has a unique solution.
- ☐ System II has no solution.
- ☐ System III has infinitely many solutions.
- ☐ Both the System II and III have infinitely many solutions.
- ☐ Both the system II and III have no solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

System I has a unique solution.
System II has no solution.
System III has infinitely many solutions.

Level 2:

Suppose PP is a $3 \times 3$3 × 3 real matrix as follows: $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$[101011001] $\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$[100010111]

The vector X_n Xn is defined by the recurrence relation $PX_{n-1} = X_n$ PXn−1 = Xn. If $X_2 = \begin{bmatrix} 7 \\ 8 \\ 3 \end{bmatrix}$[783]X2 = $\begin{bmatrix} \\ \\ \end{bmatrix}$, what is the sum of all the elements of X_0 X0?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

1 point

Consider the system of linear equations given below:

$$\begin{aligned} x - y + z &= 2 \\ x + y - z &= 3 \\ -x + y + z &= 4. \end{aligned}$$ x − y + zx + y − z−x + y + z = 2 = 3 = 4.

The system of linear equations has

[Hint: Use the relation between a system of linear equations and determinant of corresponding coefficient matrix.]

- ☐ no solution.
- ☐ infinitely many solutions.
- ☐ a unique solution.
- ☐ finitely many solutions.
- ☐ 3 dependent variable.
- ☐ 2 dependent and 1 independent variable.

No, the answer is incorrect.

Score: 0

Accepted Answers:

a unique solution.

3 dependent variable.

1 point

Consider the system of linear equations:

$$\begin{aligned} -x_1 + x_2 + 2x_3 &= 1 \\ 2x_1 + x_2 - 2x_3 &= -1 \\ 3x_2 + cx_3 &= d \end{aligned}$$ −1−x1 + x2 + 2x3 = 12x1 + x2 − 2x3 = −13x2 + cx3 = d

Choose the set of correct options.

[Hint: Recall the Gauss elimination method.]

- ☐ If $c = 2$ and $d = 1$c = 2 and d = 1, then the system has infinitely many solutions.
- ☐ If $c = 1$ and $d = 1$c = 1 and d = 1, then the system has infinitely many solutions.
- ☐ If $c = 1$ and $d = 2$c = 1 and d = 2, then the system has no solution.
- ☐ If $c = 3$ and $d = 2$c = 3 and d = 2, then the system has a unique solution.

No, the answer is incorrect.

Score: 0

Accepted Answers: